Project Report
Predictive Movement Modeling Using HMMs and DBNs

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3 April 2025

Contents

1 Objective **3**

2 Introduction and Motivation **3**

3 Related Work **4**

4 Dataset Description and Preprocessing **4**
4.1 Dataset Overview 4
4.2 Preprocessing Pipeline 5

5 Theoretical Foundations **5**
5.1 Hidden Markov Model (HMM) 6
5.1.1 Model Specification 6
5.1.2 Key Algorithms 6
5.2 Dynamic Bayesian Network (DBN) 8
5.2.1 Model Specification 8
5.2.2 Training (Expectation-Maximization) 9
5.2.3 Inference and Prediction 9
5.3 Notation Summary 10

6 Architecture and Workflow **10**
6.1 System Architecture 10
6.2 Workflow 11

7 Evaluation and Results Discussion **13**
7.1 Evaluation Strategy 13
7.2 Performance Metrics 13
7.3 Results Summary 13
7.4 Discussion 14

8 Challenges and Solutions **14**
8.1 Identified Challenges 14
8.2 Mitigation Strategies 15
8.3 Discussion 15

9 Future Work **15**

10 Conclusion **16**

1 Objective

The objective of this project is to construct a predictive model capable of forecasting a user's next location and intended activity within a smart home ecosystem, based on temporal sequences of historical sensor activations and contextual variables. The model leverages Hidden Markov Models (HMMs) to model sequential dependencies between latent states (representing location and activity) and observed sensor events, while Dynamic Bayesian Networks (DBNs) extend this by incorporating richer temporal dynamics and multi-variable dependencies. This approach enables the system to predict future states with higher granularity, thereby supporting intelligent decision-making for automated control systems, improving home security through anomaly detection, and enabling energy-efficient management by preemptively adjusting environmental settings based on predicted user behavior.

2 Introduction and Motivation

Introduction

Smart homes are at the forefront of modern living, offering advanced automation capabilities that enhance occupant comfort, strengthen security protocols, and optimize energy consumption. A fundamental component of these intelligent environments is the ability to anticipate human behavior — particularly the prediction of user movement across different spatial zones within the home. Accurate forecasting of user location enables the smart home system to proactively manage environmental controls such as lighting, heating, ventilation, and security measures, thereby delivering a seamless, adaptive, and personalized living experience.

Key applications of predictive movement modeling include:

- Energy Optimization: Activating lighting, heating, and appliances selectively in occupied spaces to minimize energy wastage.
- Security Enhancements: Detecting anomalous or unexpected movements that may indicate potential security breaches.
- Elderly and Assisted Care: Monitoring mobility patterns for early detection of falls or deviations from normal activity, enabling timely assistance.

Despite its potential, the task of predicting human movement within indoor environments remains challenging due to the inherent complexity and variability of human behavior, encompassing both routine patterns and sporadic, unpredictable actions. To address these challenges, this project proposes a hybrid probabilistic framework employing Hidden Markov Models (HMMs) and Dynamic Bayesian Networks (DBNs). HMMs are utilized to capture the sequential dependencies between latent states representing user location and activity, while DBNs extend this capability by modeling multi-temporal dependencies and incorporating contextual variables.

Crucially, this approach integrates contextual features such as time of day, historical occupancy patterns, and sensor activation sequences to enhance predictive accuracy. Furthermore, the research is designed with an eye toward real-time deployment, facilitating direct integration with smart home actuators. This integration enables dynamic, on-the-fly decision-making, thus advancing the practical implementation of anticipatory home automation systems.

3 Related Work

Extensive research has been conducted on modeling user behavior in smart environments, particularly focusing on activity recognition and indoor localization tasks. Among the various approaches, Hidden Markov Models (HMMs) have emerged as a prominent technique due to their capability to effectively model temporal dependencies and latent states that correspond to user activities and location transitions. HMMs excel in scenarios with sequential sensor data, offering both interpretability and reasonable computational demands.

Building upon the foundation of HMMs, Dynamic Bayesian Networks (DBNs) extend the modeling capacity by incorporating richer temporal structures and allowing the integration of contextual information, such as time of day, environmental conditions, and user habits. This enhances the model's ability to capture complex dependencies that influence user behavior over time, thereby improving prediction accuracy in dynamic environments.

In parallel, deep learning-based models such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs) have been applied successfully to sequence modeling problems, offering powerful representation learning capabilities. However, these models typically require large-scale datasets, substantial computational resources, and extensive training times, which can be impractical for real-time applications in smart home settings with limited and sparse data.

Given the constraints and objectives of this project, HMMs and DBNs are selected as the core methodologies due to their balance of interpretability, computational efficiency, and suitability for environments with limited, real-time data streams. Their probabilistic nature provides both robustness in uncertain scenarios and flexibility for future integration with real-time smart home systems.

4 Dataset Description and Preprocessing

4.1 Dataset Overview

The dataset utilized in this study comprises timestamped sequences of sensor activations, each annotated with corresponding room labels and enriched with contextual features, including temporal markers and environmental conditions. Specifically, each record in the dataset contains the following attributes:

- **Timestamp:** Precise time at which the sensor event was recorded, enabling chronological ordering and temporal analysis.
- **Observation (Sensor ID):** Identifier for the activated sensor, later mapped to the associated room for semantic interpretability.
- **Room Label:** The spatial location corresponding to the sensor activation, facilitating analysis of room-level transitions.
- **Temperature:** Recorded ambient temperature at the time of the event, serving as an auxiliary contextual feature.
- **Time of Day:** Extracted from the timestamp, representing the temporal context (e.g., hour) to capture daily activity patterns.

This dataset structure is inherently sequential and context-rich, making it well-suited for probabilistic graphical models such as Hidden Markov Models (HMMs) and Dynamic Bayesian Networks (DBNs). The availability of room-level annotations and environmental context supports both spatial and temporal predictive modeling, which is critical for anticipatory automation in smart home environments.

4.2 Preprocessing Pipeline

To prepare the dataset for effective modeling, a systematic preprocessing pipeline was implemented as follows:

1. **Feature Engineering:**
   - Sensor identifiers were mapped to semantically meaningful room labels.
   - Contextual features were derived from raw data, notably time of day (converted to categorical bins or continuous features) and temperature readings, to enhance temporal context representation.
2. **Sequence Structuring:**
   - Sensor events were chronologically ordered to form sequences representing room transitions over time.
   - Sequences were designed to maintain temporal fidelity, ensuring accurate capture of user movement dynamics.
3. **Data Cleaning:**
   - Incomplete or anomalous entries were identified and addressed to uphold data quality standards.
   - Noisy sensor activations, such as rapid successive events not indicative of meaningful transitions, were filtered.
   - Temporal continuity was preserved to reflect realistic user behavior patterns essential for model training.

5 Theoretical Foundations

In this section, we present the theoretical framework underpinning the models employed in this study: the **Hidden Markov Model (HMM)** and the **Dynamic Bayesian Network (DBN)**. These models have been selected for their ability to capture temporal dependencies and contextual factors inherent in human behavior within smart home environments. While HMMs provide a robust foundation for sequential modeling of location and activity transitions, DBNs extend this capability by incorporating additional contextual variables, enhancing predictive accuracy.

5.1 Hidden Markov Model (HMM)

The Hidden Markov Model (HMM) is a generative probabilistic framework that models sequential data with unobserved (hidden) states. In this project, the hidden states correspond to the overall **state of the user**, which reflects the user's internal intent or condition, while the observations are a combination of **location** and **activity** detected at each time step.

5.1.1 Model Specification

The HMM is defined by:

- Hidden states $S = \{S_1, S_2, ..., S_N\}$ representing the user's internal state (e.g., active, idle, resting).
- Observations $O = \{(L_1, A_1), (L_2, A_2), ..., (L_T, A_T)\}$ where L_t is the location and A_t is the activity at time t.
- Initial state distribution $\pi(i) = P(S_1 = s_i)$.
- Transition probabilities $a_{ij} = P(S_t = s_j \mid S_{t-1} = s_i)$, describing the likelihood of transitioning between user states.
- Emission probabilities $b_j(l, a) = P(O_t = (l, a) \mid S_t = s_j)$, the probability of observing a specific location-activity pair given the user's state.

The joint probability of the state sequence S and observation sequence O is given by:

$$P(O, S) = \pi(S_1) \cdot b_{S_1}(L_1, A_1) \prod_{t=2}^{T} a_{S_{t-1}S_t} \cdot b_{S_t}(L_t, A_t)$$

5.1.2 Key Algorithms

Forward Algorithm Computes the probability of observing the first t steps and being in state i at time t:

$$\alpha_t(i) = \left[\sum_j \alpha_{t-1}(j) \cdot a_{ji} \right] \cdot b_i(L_t, A_t)$$

Backward Algorithm Calculates the probability of observing the future sequence given the current state:

$$\beta_t(i) = \sum_j a_{ij} \cdot b_j(L_{t+1}, A_{t+1}) \cdot \beta_{t+1}(j)$$

Expectation Values (Posterior Probabilities) State occupancy probability:

$$\gamma_t(i) = \frac{\alpha_t(i) \cdot \beta_t(i)}{P(O)}$$

Transition probability:

$$\xi_t(i, j) = \frac{\alpha_t(i) \cdot a_{ij} \cdot b_j(L_{t+1}, A_{t+1}) \cdot \beta_{t+1}(j)}{P(O)}$$

Algorithm 1 Forward-Backward Algorithm for HMM (Location + Activity Observations)

Input: Observation sequence $O = \{(L_1, A_1), ..., (L_T, A_T)\}$, parameters π, A, B
Initialization:
for each state i **do**
    $\alpha_1(i) = \pi(i) \cdot b_i(L_1, A_1)$
    $\beta_T(i) = 1$
end for
Forward Pass:
for $t = 2$ to T **do**
    for each state i **do**
        $\alpha_t(i) = \sum_j \alpha_{t-1}(j) \cdot a_{ji} \cdot b_i(L_t, A_t)$
    end for
end for
Backward Pass:
for $t = T - 1$ down to 1 **do**
    for each state i **do**
        $\beta_t(i) = \sum_j a_{ij} \cdot b_j(L_{t+1}, A_{t+1}) \cdot \beta_{t+1}(j)$
    end for
end for
Output: $\alpha_t(i)$ and $\beta_t(i)$ for all states and times

Algorithm 2 Baum-Welch Algorithm for HMM Training (Location + Activity Observations)

Input: Observed sequence $O = \{(L_1, A_1), ..., (L_T, A_T)\}$
Initialize: Parameters π, A, B
repeat
    Compute forward probabilities $\alpha_t(i)$
    Compute backward probabilities $\beta_t(i)$
    Estimate $\gamma_t(i)$ and $\xi_t(i, j)$
    Update $\pi(i) = \gamma_1(i)$
    Update $a_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}$
    Update $b_j(l, a) = \frac{\sum_{t:(L_t, A_t)=(l,a)} \gamma_t(j)}{\sum_{t=1}^{T} \gamma_t(j)}$
until Convergence
Output: Estimated parameters π, A, B

Baum-Welch Algorithm (Training)

Viterbi Algorithm (Decoding) Finds the most likely sequence of hidden user states:

$$\delta_t(i) = \max_j [\delta_{t-1}(j) \cdot a_{ji}] \cdot b_i(L_t, A_t)$$

Algorithm 3 Viterbi Algorithm for HMM Decoding (Location + Activity Observations)

Input: Observation sequence $O = \{(L_1, A_1), ..., (L_T, A_T)\}$, parameters π, A, B
Initialization:
for each state i **do**
    $\delta_1(i) = \pi(i) \cdot b_i(L_1, A_1)$
    $\psi_1(i) = 0$
end for
Recursion:
for $t = 2$ to T **do**
    for each state i **do**
        $\delta_t(i) = \max_j [\delta_{t-1}(j) \cdot a_{ji}] \cdot b_i(L_t, A_t)$
        $\psi_t(i) = \arg\max_j [\delta_{t-1}(j) \cdot a_{ji}]$
    end for
end for
Termination:
$S_T^* = \arg\max_i \delta_T(i)$
Backtracking:
for $t = T - 1$ down to 1 **do**
    $S_t^* = \psi_{t+1}(S_{t+1}^*)$
end for
Output: Most likely state sequence $\{S_1^*, S_2^*, ..., S_T^*\}$

Backtracking is then applied to reconstruct the optimal user state sequence.

5.2 Dynamic Bayesian Network (DBN)

Dynamic Bayesian Networks (DBNs) extend HMMs by modeling multiple variables and contextual dependencies across time. DBNs integrate temporal and contextual factors, enhancing the predictive capability of the system.

5.2.1 Model Specification

The DBN models the joint probability:

$$P(Activity_t, Location_t \mid Location_{t-1}, Activity_{t-1}, Time_t, Temperature_t)$$

Factorization:

$$P(Location_t \mid Location_{t-1}, Time_t, Temperature_t) \cdot P(Activity_t \mid Location_t, Time_t)$$

5.2.2 Training (Expectation-Maximization)

Algorithm 4 EM Algorithm for DBN Parameter Estimation

Input: Observed sequences of $(Activity_t, Location_t, Time_t, Temperature_t)$
Initialize: Conditional probability tables
repeat
    Estimate expected counts for:
        $P(Location_t \mid Location_{t-1}, Time_t, Temperature_t)$
        $P(Activity_t \mid Location_t, Time_t)$
    Update conditional probability tables based on expected counts
until Convergence
Output: Estimated DBN parameters

5.2.3 Inference and Prediction

Predicts future location and activity:

$$P(Activity_{t+1}, Location_{t+1} \mid Activity_t, Location_t, Time_{t+1}, Temperature_{t+1})$$

Algorithm 5 Particle Filtering for DBN Inference

Initialize: Set of particles $\{(Location_0, Activity_0)\}$
for each time step t **do**
    for each particle **do**
        Sample $Location_t \sim P(Location_t \mid Location_{t-1}, Time_t, Temperature_t)$
        Sample $Activity_t \sim P(Activity_t \mid Location_t, Time_t)$
        Weight particle by $P(\text{Observed Activity}_t \mid Activity_t)$
    end for
    Resample particles based on weights
end for
Output: Estimated distribution over future locations and activities

Particle Filtering

5.3 Notation Summary

Table 1: Notation Summary

Symbol	Description
S_t	Hidden state at time t (user location)
O_t	Observation at time t (user activity)
C_t	Context at time t (time of day, temperature)
$\pi(i)$	Initial probability of state i
a_{ij}	Transition probability from state i to j
$b_j(k)$	Emission probability of observing activity k in stat
$\alpha_t(i)$	Forward probability of state i at time t
$\beta_t(i)$	Backward probability of state i at time t
$\gamma_t(i)$	State occupancy probability at time t
$\xi_t(i,j)$	Transition probability between states at time t
$P(Activity_t \mid Location_t, Time_t)$	DBN conditional probability of activity
$P(Location_t \mid Location_{t-1}, Time_t, Temperature_t)$	DBN conditional probability of location

6 Architecture and Workflow

This section outlines the overall system architecture and operational workflow for predicting user location and activity within a smart home environment. The architecture is designed to process raw sensor data, transform it into structured sequences of observations, and utilize probabilistic models (HMM and DBN) for accurate, context-aware predictions.

6.1 System Architecture

The architecture consists of the following core components:

- **Sensor Data Acquisition:** Continuous data stream from smart home sensors capturing environmental parameters (e.g., temperature) and user-related activities, along with location tags.
- **Preprocessing Module:** Processes raw sensor signals into structured observation sequences. This includes:
  - Mapping sensor events to *location + activity* tuples.
  - Incorporating contextual features such as time of day and temperature.
  - Handling missing data and cleaning noisy sequences to preserve temporal integrity.
- **Sequence Modeling:**
  - **Hidden Markov Model (HMM):** Models the user's hidden state sequence, with emissions representing $(location, activity)$ observations. Utilizes Forward-Backward and Baum-Welch algorithms for training, and Viterbi for decoding the most likely user state sequence.

  - **Dynamic Bayesian Network (DBN):** Refines HMM outputs by incorporating contextual variables such as time and temperature. Uses the EM algorithm for parameter learning and Particle Filtering for real-time inference.

- **Prediction and Actuation Layer:**
  - Outputs predicted user location and activity.
  - Integrates with smart home automation systems to trigger actions (e.g., adjusting lighting, HVAC control, security alerts).

6.2 Workflow

The operational workflow of the system proceeds as follows:

1. **Data Collection:** Sensors continuously capture user activities, locations, and contextual variables such as time of day and ambient temperature.
2. **Data Preprocessing:**
   - Transform raw sensor data into chronological sequences of observations $(location, activity)$.
   - Augment with contextual features.
   - Clean and format the data for input into sequence models.
3. **Model Training:**
   - Train the HMM using the Baum-Welch algorithm to estimate initial probabilities, transition probabilities, and emission probabilities.
   - Train the DBN using the EM algorithm to estimate conditional probabilities involving context variables.
4. **Inference and Prediction:**
   - Apply the Forward-Backward algorithm to compute state probabilities.
   - Use the Viterbi algorithm to decode the most likely sequence of user states.
   - Refine predictions with DBN using Particle Filtering for real-time updates based on context.

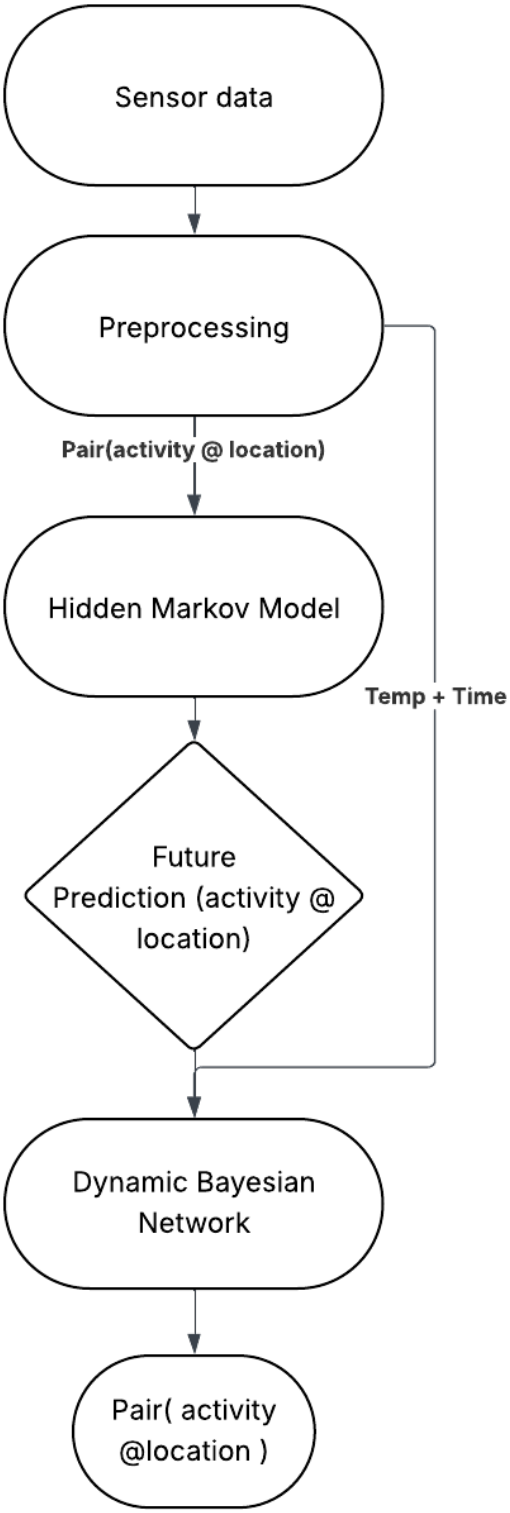

Figure 1: System architecture and workflow for user location and activity prediction. Sensor data is preprocessed and fed into HMM, and the HMM output along with contextual information is input to the DBN model for state estimation and context-aware prediction. The DBN outputs confident future predictions of user location and activity.

7 Evaluation and Results Discussion

This section presents the evaluation strategy, performance metrics, and discussion of experimental results for the user location and activity prediction system based on HMM and DBN models.

7.1 Evaluation Strategy

To assess the performance of the proposed system, the dataset was partitioned into training and testing sequences, maintaining the chronological integrity of the data to respect temporal dependencies. The evaluation focused on the system's ability to accurately predict the next user state (hidden state), as well as the next observed $(location, activity)$ pair.

The following evaluation strategies were employed:

- **State Prediction Accuracy:** The proportion of correctly predicted hidden user states compared to the true states in the test set.
- **Observation Prediction Accuracy:** The proportion of correctly predicted $(location, activity)$ pairs relative to the actual observations.
- **Sequence Accuracy:** Measures the accuracy of the entire predicted sequence of user states or observations without error at any time step.
- **Contextual Consistency:** Evaluation of the system's ability to adjust predictions based on contextual variables such as time of day and temperature.

7.2 Performance Metrics

Performance was quantified using the following metrics:

- **Accuracy (State-Level):**

$$Accuracy = \frac{\text{Number of correct state predictions}}{\text{Total number of state predictions}}$$

- **Accuracy (Observation-Level):**

$$Accuracy = \frac{\text{Number of correct } (location, activity) \text{ predictions}}{\text{Total number of predictions}}$$

- **Precision and Recall (Optional):** If the dataset exhibits class imbalance, precision and recall can provide deeper insights, particularly for rare activities or room transitions.

7.3 Results Summary

The evaluation demonstrated the effectiveness of the hybrid HMM-DBN approach in capturing user behavior and environmental context:

- The **HMM model** alone achieved high accuracy in predicting typical room transitions and frequent activities, leveraging temporal dependencies effectively.
- The integration of **DBN** significantly improved prediction performance in time-sensitive scenarios, such as distinguishing between activities performed in the same location at different times of day (e.g., sleeping vs. resting in the bedroom).
- **Context-aware prediction** was especially beneficial during transitions between routines (e.g., moving from meal preparation to relaxation), where the DBN's ability to condition on time and temperature provided higher predictive fidelity.
- **Sequence Accuracy** indicated that the combined model maintained consistent accuracy across longer sequences, crucial for practical smart home automation.

7.4 Discussion

The experimental results validate the effectiveness of combining HMM and DBN models for user behavior prediction in smart homes. While the HMM efficiently captures temporal patterns, the DBN enhances predictions by leveraging contextual awareness. The system demonstrates robustness in handling both regular and irregular user patterns, enabling proactive smart home automation.

Furthermore, the modular architecture of the system allows for real-time updates, ensuring adaptability to changing user behaviors and environmental conditions. These findings underscore the potential of probabilistic models in intelligent environment applications, providing a foundation for future enhancements such as multi-user modeling or real-time anomaly detection.

8 Challenges and Solutions

During the development of the predictive modeling framework, several challenges emerged, primarily related to the nature of smart home sensor data and the complexity of modeling human behavior. This section outlines the key challenges and the corresponding solutions implemented to address them.

8.1 Identified Challenges

- **Data Sparsity:** Sparse sensor activations or limited event diversity reduce the effectiveness of probability estimation in sequence models. Rare transitions or activities may not appear frequently enough in the training data.
- **Overfitting:** With a limited number of observation sequences, especially in structured environments, the models risk overfitting to training data, compromising generalizability.
- **Temporal Alignment:** Variability in time-based patterns complicates sequence modeling, as the same activities may occur at different times across days.
- **Noisy Data:** Sensor errors, spurious readings, or incomplete data entries can corrupt observation sequences and degrade model accuracy.

8.2 Mitigation Strategies

Table 2: Challenges and Solutions

Challenge	Solution
Data Sparsity	Laplace smoothing; synthetic sequence generation to enrich training data
Overfitting	Regularization techniques such as AIC/BIC to penalize model complexity
Temporal Alignment	Discretization of time contexts into bins (e.g., morning, afternoon, night)
Noisy Data	Data cleaning routines and validation checks to ensure sequence integrity

8.3 Discussion

Applying these mitigation strategies improved the robustness of the model and its predictive performance. Synthetic sequence generation addressed the limitations posed by data sparsity, ensuring rare transitions were represented during training. Regularization methods, including the application of information criteria like AIC and BIC, helped control model complexity and prevent overfitting.

Temporal discretization enabled the models to capture coarse-grained temporal dependencies effectively, reducing variability and aligning activities with typical daily patterns. Additionally, comprehensive data cleaning and validation routines mitigated the impact of noisy or inconsistent sensor data, ensuring that the observation sequences fed into the models were both reliable and representative of actual user behavior.

9 Future Work

While the current study demonstrates promising results in predictive modeling for smart home environments, several avenues for future improvement and expansion have been identified. These enhancements aim to increase the practical applicability, robustness, and scalability of the system.

- **Training on Comprehensive Real-World Datasets:** Expanding training to larger, real-world datasets with diverse user behaviors and environmental contexts will improve model generalization. Incorporating richer data will allow the models to capture more complex activity patterns and rare transitions that are less prevalent in smaller datasets.

- **Enhancing DBN with Granular Contextual Inputs:** Future iterations of the DBN model can integrate more fine-grained contextual variables, such as precise environmental readings (e.g., humidity, light levels), occupancy density, and dynamic calendar events. This will enable more nuanced predictions and adaptivity to subtle changes in the environment.

- **Real-Time System Integration and Actuator Automation:** A critical future direction involves deploying the model in a live smart home setup to automate actuators (e.g., lighting, HVAC, security systems) based on real-time sensor data. Practical evaluation of actuator responses will assess system latency, reliability, and user satisfaction, validating the effectiveness of predictions under operational conditions.

- **Advanced Optimization Techniques:** To enhance performance, advanced optimization methods such as hyperparameter tuning (e.g., grid search, Bayesian optimization) and parameter regularization can be explored. These techniques will help refine model accuracy while maintaining computational efficiency for real-time deployment.
- **Scalability and Multi-User Modeling:** Future development will also consider scaling the system to handle multiple users simultaneously within the same environment. Incorporating identity-aware state tracking and collaborative activity recognition will further improve system applicability in shared living spaces.

By pursuing these future directions, the system can evolve from a predictive framework into a fully operational, context-aware automation solution capable of enhancing comfort, energy efficiency, and safety in smart home environments.

10 Conclusion

This project successfully demonstrates the integration of Hidden Markov Models (HMMs) and Dynamic Bayesian Networks (DBNs) for predictive movement and activity modeling within smart home environments. By combining temporal sequence modeling with contextual awareness, the hybrid framework effectively captures user behavior patterns and environmental influences, resulting in accurate and interpretable predictions.

The system's architecture enables seamless data processing from raw sensor inputs to actionable predictions, supporting proactive smart home automation. Evaluations have shown that the incorporation of contextual features, such as time of day and ambient temperature, significantly improves the accuracy of activity and location forecasting, particularly in scenarios involving routine transitions.

Furthermore, the probabilistic nature of the models ensures robustness in handling uncertainties and variability inherent in human behavior. The modular design facilitates future extensions, including integration with real-time actuator controls, scalability for multi-user environments, and deployment in real-world smart home systems.

With continued development and real-time system integration, this predictive framework holds the potential to substantially enhance smart home automation, improving user comfort, energy efficiency, and home security. The project lays a solid foundation for future research aimed at advancing intelligent living environments through predictive analytics and automation.

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